Sanghyun Lee

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Research Interests

My research is rooted in diffusion language models, with a current focus on discrete diffusion models and unified multimodal models. The question that connects my work is how model design should adapt to each modeling paradigm and modality: rather than adopting components developed for autoregressive models, I study architectures, tokenization, and representations that build on the properties of the diffusion process itself.

On the training side, I am interested in efficient and scalable training methods for new modeling paradigms, including billion-scale pretraining from scratch, and in understanding when a shared generative formulation creates synergy across language and vision. On the inference side, I am interested in test-time scaling, designing inference algorithms that effectively scale computation at generation time.

Experience

Krafton 2025.06 – 2026.04
Research Intern, Foundation Research Team *Seoul, South Korea*

Education

Korea Advanced Institute of Science and Technology (KAIST) 2025.09 – Present
Integrated M.S.-Ph.D., Kim Jaechul Graduate School of AI *Daejeon, South Korea*
Advisor: Prof. Seungryong Kim

Korea Advanced Institute of Science and Technology (KAIST) 2019.03 – 2025.09
B.S. in Electronic Engineering, Minor in Mathematical Sciences *Daejeon, South Korea*

Publications

Preprints

1. **"Rethinking Tokenization for Discrete Diffusion Language Models"**
   Under review. arXiv preprint coming soon.
2. **"Exploring Unified Latent Diffusion for Multimodal Modeling"**
   Under review. arXiv preprint coming soon.
3. **Sanghyun Lee***, Wonjin Nam*, Seungho Jang, Jaeho Lee, Jiho Park, Jaewoon Lee, Seungryong Kim. **"ReAnchor: Reusing Rejected Verifier States for Diffusion Drafting"**
   arXiv preprint, 2026. Under review. [link]

International Conference

1. **Sanghyun Lee**, Chunsan Hong, Seungryong Kim, Jonghyun Lee, Jongho Park, Dongmin Park. **"Looped Diffusion Language Models"**
   Advances in Neural Information Processing Systems (NeurIPS), 2026. [link]
   Conference on Language Modeling (COLM) workshop, 2026. Spotlight Paper.
2. Chunsan Hong*, **Sanghyun Lee***, Chieh-Hsin Lai, S. Hayakawa, Y. Takida, Yuki Mitsufuji, Seungryong Kim, J.C. Ye. **"Understanding and Accelerating the Training of Masked Diffusion Language Models"**
   International Conference on Machine Learning (ICML) workshop, 2026. [link]
3. **Sanghyun Lee**, Seungryong Kim, Jongho Park, Dongmin Park. **"Lookahead Unmasking Elicits Accurate Decoding in Diffusion Language Models"**
   International Conference on Machine Learning (ICML), 2026. [link]
4. Chunsan Hong, **Sanghyun Lee**, J.C. Ye. **"Unifying Masked Diffusion Models with Various Generation Orders and Beyond"**
   International Conference on Machine Learning (ICML), 2026. Spotlight Paper (Acceptance rate < 2.2%). [link]

5. **Sanghyun Lee**, Sunwoo Kim, Seungryong Kim, Jongho Park, Dongmin Park. **"Effective Test-Time Scaling of Discrete Diffusion through Iterative Refinement"**
International Conference on Machine Learning (ICML) workshop, 2026. [link]

6. Donghoon Ahn*, Jiwon Kang*, **Sanghyun Lee**, Jaewon Min, Minjae Kim, Wooseok Jang, Hyoungwon Cho, Sayak Paul, SeonHwa Kim, Eunju Cha, Kyong Hwan Jin, Seungryong Kim. **"A Noise Is Worth Diffusion Guidance"**
International Conference on Learning Representations (ICLR), 2026. [link]

7. Donghoon Ahn*, Jiwon Kang*, **Sanghyun Lee**, Minjae Kim, Jaewon Min, Wooseok Jang, Sangwu Lee, Sayak Paul, Susung Hong, Seungryong Kim. **"Fine-Grained Perturbation Guidance via Attention Head Selection"**
Advances in Neural Information Processing Systems (NeurIPS), 2025. [link]

Research Areas

- Discrete diffusion models
- Unified multimodal models
- Test-time scaling
- Probabilistic modeling
- Efficient from-scratch pretraining

Skills

- **Programming and Systems:** Python, PyTorch, C/C++, CUDA
- **Large-Scale Training Experience:**
  - Billion-scale language model and unified multimodal model from-scratch pretraining
  - Training experience on multi-GPU clusters
- **Academic Knowledge:** Diffusion Models, Reinforcement Learning Theory, Information Theory, Linear Algebra, MC Methods, Optimization Theory, Measure Theory, Real Analysis